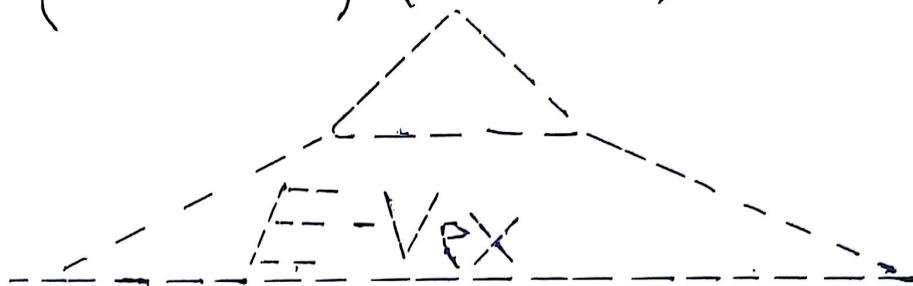


$$y = e^{4x^3 - 5x}$$

$$\Rightarrow y' = (e^{4x^3 - 5x}) \cdot (12x^2 - 5)$$



$$y = \ln(\tan(x))$$

$$\Rightarrow y' = \left(\frac{1}{\tan(x)} \right) (\sec^2 x)$$

Simplify:

$$y' = \frac{\sec^2 x}{\tan(x)} \quad \text{and} \quad y' = \left(\frac{\cos(x)}{\sin(x)} \right) \left(\frac{1}{\cos^2 x} \right)$$

$$\Rightarrow \frac{\cancel{1}}{\sin(x)\cos(x)} = y'$$